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Week 3: Uncalibrated Stereo

✓ **Ungraded Plugin:** 3.1
Overview of Uncalibrated Stereo
6 min

✓ **Ungraded Plugin:** 3.2
Problem of Uncalibrated Stereo
6 min

✓ **Practice Assignment:** 3.2
Problem of Uncalibrated Stereo Self-Check Quiz
Submitted

✓ **Ungraded Plugin:** 3.3
Epipolar Geometry
15 min

✓ **Practice Assignment:** 3.3
Epipolar Geometry Self-Check Quiz
Submitted

✓ **Ungraded Plugin:** 3.4
Stereo Vision in Nature
14 min

✓ **Practice Assignment:** 3.4
Stereo Vision in Nature Self-Check Quiz
Submitted

✓ **Reading:** 3.5 Video Correction
10 min

⏪ **Ungraded Plugin:** 3.5
Estimating Fundamental Matrix
6 min

📋 **Practice Assignment:** 3.5
Estimating Fundamental Matrix Self-Check Quiz
3h

⏪ **Ungraded Plugin:** 3.6
Finding Correspondences
7 min

📋 **Practice Assignment:** 3.6
Finding Correspondences Self-Check Quiz
3h

📖 **Reading:** 3.7 Video Correction
10 min

⏪ **Ungraded Plugin:** 3.7
Computing Depth
10 min

📋 **Practice Assignment:** 3.7
Computing Depth Self-Check Quiz
3h

🗨️ **Discussion Prompt:** Week 3 Simple Binocular Stereo System
10 min

🗨️ **Discussion Prompt:** Week 3 Questions and Feedback

Week 3 Quiz

3.5 Video Correction

In [Video 3.5 Estimating Fundamental Matrix](#), from 1:43 to 2:17, the first three terms in the f matrix should be f_{11} , f_{12} , and f_{13} instead of f_{11} , f_{21} , and f_{31} . In the second image, we also made some corresponding corrections (see below).

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

Stereo Calibration Procedure

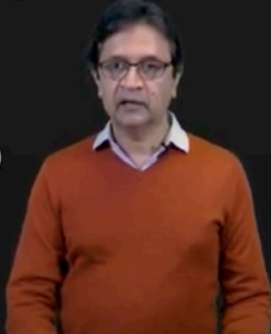
Step B: Rearrange terms to form a linear system.

$$\begin{bmatrix} u_l^{(1)}u_r^{(1)} & u_l^{(1)}v_r^{(1)} & u_l^{(1)} & v_l^{(1)}u_r^{(1)} & v_l^{(1)}v_r^{(1)} & v_l^{(1)} & u_r^{(1)} & v_r^{(1)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(i)}u_r^{(i)} & u_l^{(i)}v_r^{(i)} & u_l^{(i)} & v_l^{(i)}u_r^{(i)} & v_l^{(i)}v_r^{(i)} & v_l^{(i)} & u_r^{(i)} & v_r^{(i)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(m)}u_r^{(m)} & u_l^{(m)}v_r^{(m)} & u_l^{(m)} & v_l^{(m)}u_r^{(m)} & v_l^{(m)}v_r^{(m)} & v_l^{(m)} & u_r^{(m)} & v_r^{(m)} & 1 \end{bmatrix}$$

A
(Known)

$$\begin{bmatrix} f_{11} \\ f_{21} \\ f_{31} \\ f_{22} \\ f_{23} \\ f_{32} \\ f_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

f
(Unknown)



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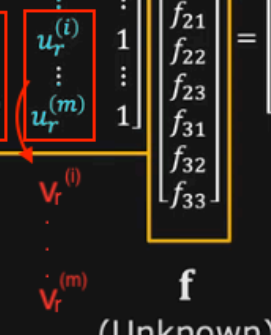
Step B: Rearrange terms to form a linear system.

$$\begin{bmatrix} u_l^{(1)}u_r^{(1)} & u_l^{(1)}v_r^{(1)} & u_l^{(1)} & v_l^{(1)}u_r^{(1)} & v_l^{(1)}v_r^{(1)} & v_l^{(1)} & u_r^{(1)} & v_r^{(1)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(i)}u_r^{(i)} & u_l^{(i)}v_r^{(i)} & u_l^{(i)} & v_l^{(i)}u_r^{(i)} & v_l^{(i)}v_r^{(i)} & v_l^{(i)} & u_r^{(i)} & v_r^{(i)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_l^{(m)}u_r^{(m)} & u_l^{(m)}v_r^{(m)} & u_l^{(m)} & v_l^{(m)}u_r^{(m)} & v_l^{(m)}v_r^{(m)} & v_l^{(m)} & u_r^{(m)} & v_r^{(m)} & 1 \end{bmatrix}$$

A
(Known)

$$\begin{bmatrix} f_{11} \\ f_{21} \\ f_{31} \\ f_{22} \\ f_{23} \\ f_{32} \\ f_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

f
(Unknown)



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